

SILICONE RUBBER INSULATOR COATINGS

GENERAL

Insulator coating materials must be of room temperature vulcanising (RTV) silicone rubber formulation, specifically developed for use on outdoor high voltage porcelain, glass, resin and rubber insulation. The coating must be easily applied using standard spray-painting equipment and yield a naturally smooth finish.

CHARACTERISTICS

In addition to being highly hydrophobic, other necessary design characteristics are:

- fast hydrophobicity transfer to contaminating layers
- fast recovery of hydrophobicity after partial discharge activity
- excellent arc resistance
- resistance to damage by corona discharge
- resistance to deterioration from ultraviolet (UV) radiation
- resistance to degradation by atmospheric and chemical contaminants such as marine salts and industrial pollutants
- good adhesion to glass, ceramic and resin surfaces

PROPERTIES

The coating should possess the properties specified in the table below.

PROPERTY	TEST METHOD	UNIT	VALUE
Specific gravity @ 23°C			0,9 to 1.25
Dielectric strength @ 23°C	BS 2918	kV/mm	>50
Surface resistivity	IEC 60003	ohm	$>1.25 \times 10^{13}$
Volume resistivity	IEC 60093	ohm metre	$>2.25 \times 10^{13}$
Hydrophobicity	IEC 62073	contact angle	>90°
Arc resistance	ASTM D 495-99	seconds	>185
Tracking and Erosion	ASTM D 2303-97	minutes @ 2.5kV	>470
Hydrophobicity Recovery	IEC 62073 100 hrs corona	Wettability class: HC1 - HC6	HC1-HC2 within 48 hrs
Adhesion	ISO 16276-2		Class 1
Adhesion	IEC 62217 100 hrs boiling		No loss of adhesion
Skin-over time 23° C		minutes	<25

COMPOSITION

In view of the more controlled curing rate and the extended shelf life, two-part silicone rubber compositions are preferred. The shelf life, when stored at temperatures between 5°C and 30°C must be a minimum of 12 months.

After curing, the silicone rubber content must be more than 50% by weight.

The filler used to ensure partial arc resistance must be alumina trihydrate.

The coating must be supplied with a suitable primer to ensure excellent adhesion to glass, ceramic, resin and rubber substrates.

PERFORMANCE

The material must have displayed satisfactory and maintenance-free service performance over a period at least 10 years at sites of at least “Very Heavy” IEC pollution severity classification.

In addition to the above, the coating must have passed a performance evaluation of a minimum of one year duration at a naturally polluted insulator test station as per the Eskom Procedure DPC 34-224 or similar.

PACKING

The material must be available in 1 litre and 5 litre tins accompanied by pre-measured volumes of catalyst and primer. The tins must be marked with a batch number and the manufacturing date.